Engineered Science

Heart Disease Risk Prediction Using Deep Learning and Probabilistic Calibration Techniques

Journal:	*Engineered Science*
Manuscript ID	ES-2026-0082.R2
Manuscript Type:	Original Research
Date Submitted by the Author:	18-May-2026
Complete List of Authors:	VENKATESH, BAJJAREDDY; Amrita Vishwa Vidyapeetham Amrita School of Engineering Bengaluru, Mathematics M R, Rashmi; Amrita Vishwa Vidyapeetham Amrita School of Engineering Bengaluru, Dept. of Electrical and Electronics Engineering K, MEERA; Amrita Vishwa Vidyapeetham Amrita School of Engineering Bengaluru, Mathematics Lee , Hoong Pin; INTI International University, Nilai, Malaysia, Faculty of Engineering and QS Hima Sagar , B; Amrita Vishwa Vidyapeetham Amrita School of Engineering Bengaluru, Dept. of Computer Science and Engineering A, Vibhas Reddy; Amrita Vishwa Vidyapeetham Amrita School of Engineering Bengaluru, Computer Science Irigela, Bharghav Vardhan Reddy; Amrita Vishwa Vidyapeetham Amrita School of Engineering Bengalore, Dept. of computer Science and Engineering
Keywords:	Bayesian Bootstrap Aggregation, Heart Attack Prediction, Multi Layer Perceptron, Probabilistic Calibration, Platt Scaling

SCHOLARONE MANUSCRIPTS

From
Dr. B. Venkatesh
Professor
Department of Mathematics
Amrita School of Engineering, Bangalore-560035
Amrita Vishwa Vidyapeetham, India

To
The Editor-in -Chief
Engineered Science Journal

Dated: 20.02.2026

Subject: Submission of Manuscript Titled “Heart Disease Risk Prediction Using Deep Learning and Probabilistic Calibration Techniques”

Dear Editor, I am pleased to submit our manuscript titled “Heart Disease Risk Prediction Using Deep Learning and Probabilistic Calibration Techniques” for consideration for publication in the Engineered Science Journal. This paper provides a good foundation for a machine learning model that incorporates deep learning and probabilistic calibration to improve the ac curacy and reliability of predictions regarding heart attacks. We confirm that the manuscript is original, has not been published before, and is not currently being considered for publication elsewhere. All authors have read and approved the manuscript and have no competing interests to disclose. Thank you for considering our manuscript for publication. We look forward to your positive response.

Sincerely,
B. Venkatesh
Corresponding Author

For Review Only

20-word Innovative Description

This paper provides a good foundation for a machine learning model that incorporates deep learning and probabilistic calibration to improve the ac curacy and reliability of predictions regarding heart attacks.

For Review Only

Table of Content



Heart Disease Risk Prediction Using Deep Learning and Probabilistic Calibration Techniques

Venkatesh B[1]*, Rashmi M. R[2], K. N. Meera[1], Lee Hoong Pin[3], Hima Sagar B.[4], A. Vibhas Reddy[4], Irigela Bharghav Vardhan Reddy[4]

[1] Department of Mathematics, Amrita School of Engineering, Bengaluru -560035, Amrita Vishwa Vidyapeetham, India
[2] Department of Electrical and Electronics Engineering, Amrita School of Engineering, Bengaluru- 560035, Amrita Vishwa Vidyapeetham, India
[3] Faculty of Engineering and QS, Director of the Centre for Sustainable Engineering Solutions, INTI International University, Nilai,71800, Malaysia
[4] Department of Computer Science and Engineering, Amrita School of Computing, Bengaluru -560035, Amrita Vishwa Vidyapeetham, India

For Review Only

Abstract

Heart disease, particularly heart attacks, ranks at the top of the list of causes of death worldwide. Annually, 4-5 million people succumb to heart attacks worldwide. To intervene and reduce the risk of a fatal outcome, it is necessary to have precise predictions. This paper provides a foundation for a machine learning model that incorporates deep learning and probabilistic calibration to improve the accuracy and reliability of predictions regarding heart attacks. Feature engineering approaches like outlier clipping and interaction term generation are used to preprocess the data. The system is further enhanced by using advanced calibration techniques such as Platt scaling, isotonic regression, and histogram binning, as well as ensemble approaches like bagging, using a Multi-Layer Perceptron (MLP) trained on a medically enriched dataset. Bayesian bootstrap aggregation is employed to reduce the variance and to capture the epistemic uncertainty. Evaluation metrics such as accuracy, Precision, Brier score, Recall, False Negative Rate(FNR), F1 Score, and log-loss are reported. The results of the proposed method are compared with those of the TabNet, DeepFM, XGBoost, Random Forest, and Logistic Regression algorithms. In addition to better classification accuracy, the final ensemble model offers well-calibrated risk probabilities, offering greater trust and interpretability in clinical decision-making via an interactive Streamlit-based web interface.

Keyword- Bayesian Bootstrap Aggregation, Heart Attack Prediction, Histogram Binning, Isotonic Regression, Multi Layer Perceptron (MLP), Probabilistic Calibration, Platt Scaling, Streamlit.

1 Introduction

Cardiovascular Diseases (CVDs), especially myocardial infarctions (heart attack), continue to be the number one cause of death worldwide [1]. Early and accurate risk assessment remains truly influential in prevention and therapeutic intervention [2]. Post-pandemic death rates due to heart attack are becoming high [3]. Early prediction of cardiovascular diseases can save millions of people every year [4]-[5].

In recent years, there has been much interest in the use of machine learning and deep learning algorithms for clinical decision support systems [6]-[8]. Many researchers have proposed deep learning models for heart attack prediction. Optimization algorithms were used in [9] to tune the hyperparameters of the regression models used for the prediction of heart disease. The data related to cardiovascular diseases is obtained from cardiac imaging and electrocardiogram signals, which are highly unstructured. Ensemble methods are

*Email:b_venkatesh@blr.amrita.edu

efficient for structured data. The review presented in [10] revealed that the deep learning models' performance is better for unstructured data. Though the deep learning models' performance is convincing, the clinical validation of the same is not much reported in the literature [11].

Deep learning models exhibit certain limitations, such as poor feature extraction for unbalanced classes, and are less interpretable [12]-[13]. To overcome these limitations, a preprocessing pipeline with a Deep Residual Network was proposed in [14] for heart disease prediction. The heart disease risk prediction model's reliability can be improved using post-hoc calibration beyond the discrimination metrics [15]. Streamlit, a cloud-based platform, was recommended by many researchers for computer-aided disease prediction [16]-[17]. Deep learning architecture with image restoration technique was proposed in [18] for disease recognition in tomato leaves. Transformer-based attention modules were proposed in [19] to control Nitrogen Oxide (NOx) emissions. Long Short-Term Memory Networks were used in [20] to address cybersecurity threats, including ransomware. Deep learning models were proposed in [21]-[22] for sugarcane disease prediction. This was also used to enhance the yield. A CNN algorithm was proposed in [23] for knee joint diagnosis. A combination of dynamic optimization technique with CNN was proposed in [24] for supply chain management. Xception and DenseNet201 models were proposed in [25] for skin disease classification.

The present work develops an advanced web-based heart attack risk prediction system using Streamlit, in which three MLP models are bagged together and provide calibrated predicted probabilities. Calibration techniques used include Platt scaling, isotonic regression, and histogram binning. Calibrated probabilities are vital to maintain interpretability and reliability, particularly in life-critical areas such as healthcare. Unlike standard models, which often do not produce well-calibrated probability estimates, this system takes advantage of recent advances in calibrated deep learning to produce more than just 'yes' or 'No' predictions. It provides reliable risk probabilities that are critical in clinical decision-making. The results are compared with TabNet, DeepFM, XGBoost, Random Forest, and Logistic Regression algorithms. The proposed model is found to be more reliable compared to the other models.

2 Methodology

The objective of the proposed work is to enhance the accuracy and confidence level at which it is possible to forecast heart disease by utilizing a combination of logistic regression-based feature selection and deep learning ensemble methods. The importance of this project can be understood in the context of diagnosing the disease. The next section explains the process that is adopted in this work.

2.1 Data Collection

The data for the series is procured from real-world heart-attack-prediction records, with 1753 entries and 26 key features consisting of clinical and lifestyle factors. The dataset is obtained from Kaggale. It is a curated dataset designed for research and educational purposes and does not originate from direct clinical data collection by the authors.

- The dataset consists of 1753 entries and 26 features, including demographic, lifestyle, and medical attributes related to heart attack risk.
- It was created and shared by a third party on Kaggle; the authors did not conduct any primary data collection or patient interaction.
- Ethical approval was not required as the dataset is publicly accessible and does not involve human subject experimentation.
- The dataset contains no Personally Identifiable Information (PII), such as names or contact details.
- All data is anonymized or synthetic, ensuring patient privacy and confidentiality.

Unlike artificially constructed ones such as the prestigious UCI Cleveland dataset, the collected dataset does not limit itself to basic medical variables: it goes further by including important modern lifestyle and behavioral attributes like income, stress level, hours of sleep, and frequency of physical activities. In addition, indicators for diabetes, smoking, and cardiovascular disease by way of family history are included.

For Review Only

This encompasses all the cardiovascular risk factors. Though geography-based features such as Country, Continent, and Hemisphere are included, their relevance to clinical heart attack risk is limited. These features are retained for exploratory purposes, but caution is advised when interpreting their predictive contribution. These attributes have been utilized for the training and testing of the suggested hybrid prediction model. The attributes of the dataset are provided in Table 1.

Table 1: Description of Dataset

Feature Name	**Description**	**Type**
Patient ID	Unique identifier for each patient	Categorical
Age	Age of the patient	Numerical
Sex	Gender of the patient	Categorical
Cholesterol	Cholesterol level (mg/dL)	Numerical
Systolic Pressure	Pressure during heart's contraction phase	Numerical
Diastolic Pressure	Pressure during heart's relaxation phase	Numerical
Heart Rate	Resting heart rate (bpm)	Numerical
Diabetes	Diabetes (1: Yes, 0: No)	Binary
Family History	Family history of heart disease	Binary
Smoking	Smoking status	Binary
Obesity	Obesity status	Binary
Alcohol Consumption	Alcohol use status	Binary
Exercise Hours Per Week	Weekly exercise hours	Numerical
Diet	Dietary pattern	Categorical
Previous Heart Problems	History of heart issues	Binary
Medication Use	Heart-related medication usage	Binary
Stress Level	Self-reported stress level	Numerical
Sedentary Hours Per Day	Daily sedentary hours	Numerical
Income	Annual income	Numerical
BMI	Body Mass Index	Numerical
Triglycerides	Blood triglyceride level	Numerical
Physical Activity Days Per Week	Active days per week	Numerical
Sleep Hours Per Day	Daily sleep hours	Numerical
Country	Country of residence	Categorical
Continent	Continent of residence	Categorical
Hemisphere	Hemisphere of residence	Categorical

2.2 Feature Engineering and Preprocessing

The proposed system incorporates medically inspired preprocessing steps to enhance model generalizability and interpretability. Outliers are clipped to clinically plausible ranges. For example, cholesterol is clipped between 100–500 mg/dL, and triglycerides between 30–1000 mg/dL.

Additionally, the following interaction features are engineered:

- `Cholesterol_BMI` = Cholesterol × BMI
- `BP_Ratio` = Systolic_BP / (Diastolic_BP +1)

For Review Only

- `Age_Stress` = Age × Stress Level

The feature engineering process utilized a medically inspired approach to capture synergistic effects between independent risk factors that a standard linear model might overlook.

- **Cholesterol × BMI:** This interaction models the compound metabolic risk where high lipid levels combined with obesity significantly escalate the probability of arterial plaque formation beyond the risk posed by either factor alone.
- **BP Ratio (Systolic / Diastolic + 1):** This feature serves as a proxy for pulse pressure and vascular resistance. A high ratio often indicates arterial stiffness, which is a more sensitive clinical predictor for cardiovascular events than isolated blood pressure readings.
- **Age × Stress Level:** This captures the biological vulnerability of the cardiovascular system to chronic psychological strain as a patient ages. It reflects the clinical observation that the physiological impact of high stress is more detrimental to cardiac health in older demographics.

Numerical features are normalized with `StandardScaler`, while categorical features are labeled and one-hot encoded. Class imbalance is tackled with `RandomOverSampler`.

2.3 Deep Learning Model Architecture

The predictive engine is based on a deep neural network that is implemented using a Multilayer Perceptron (MLP). This MLP is the base estimator within a bagging ensemble and models highly complex, nonlinear interactions established between multiple health features. The MLP consists of a feedforward architecture with two hidden layers of 64 and 32 neurons, respectively, each having Rectified Linear Unit (ReLU) activation function which is given in (1) [14].

$$f(x) = \max(0, x) \tag{1}$$

The selection of the Multilayer Perceptron (MLP) architecture, featuring two hidden layers of 64 and 32 neurons, was informed by both established deep learning practices and dataset-specific performance tuning.

- **Complexity Balancing:** The architecture was chosen to model highly complex, nonlinear interactions between the 26 key clinical and lifestyle features without inducing overfitting. Using a feedforward structure with decreasing layer sizes (64 to 32) acts as a feature compressor, aiding in generalized representation.
- **Hyperparameter Tuning & Stability:** The configuration was determined through careful tuning of several hyperparameters to ensure stability and generalization. These included:
  - **Activation Function:** The use of ReLU across hidden layers was selected to alleviate the vanishing gradient problem and ensure faster convergence.
  - **Regularization:** An L2 weight decay ($\alpha = 0.0005$) was implemented specifically to prevent overfitting in this high-capacity model.
  - **Optimization:** The Adam optimizer was utilized for its adaptive learning rate capabilities, tracking the first and second moments of the gradient to refine the model's performance.
  - **Ensemble Integration:** The architecture's performance was further validated by its role as a base estimator within a Bagging ensemble. This approach reduced the variance typical of individual MLPs, ensuring the 64-32 neuron structure remained resilient even with small or unbalanced medical datasets

ReLU helps alleviate the vanishing gradient problem and enables faster convergence during training. Backpropagation is used for network training with gradient descent optimization.

For Review Only

2.3.1 Hyperparameters and Training Dynamics

To ensure stability and generalization, several hyperparameters are carefully tuned. The process used for hyperparameter tuning is given below:

- **Systematic Selection Process:** Rather than arbitrary manual selection, these values were identified by monitoring the model's generalization error using an internal validation fraction during the training phase.
- **Optimization Algorithm:** The tuning process was centered around the Adam optimizer, which inherently employs adaptive learning rates by tracking the first and second moments of the gradient to refine weight updates.
- **Learning Rate:** 0.001- ensuring moderate step sizes during weight updates.
- **Regularization and Stability:** The L2 weight decay (α) was specifically tuned to 0.0005 to maintain a balance between model capacity and generalization, ensuring the MLP remained resilient to noise in the clinical dataset.
- **Early Stopping Integration:** To supplement the hyperparameter search, an early stopping mechanism was enabled. This automated process halted training once the validation performance stabilized, ensuring the selected learning rate and batch size achieved the optimal point of convergence without inducing overtraining.
- **Validation Strategy:** The robustness of these parameters was further verified using 3-fold cross-validation for the Platt scaling and isotonic regression components, ensuring the configuration was not over-fitted to a single data split.

The model is trained using Stochastic Gradient Descent (SGD) with adaptive learning, which is standard for the `MLPClassifier` used by the Python library scikit-learn. In addition, it uses a validation fraction to monitor the model's generalization. The `MLPClassifier` uses the Adam optimizer, a type of SGD that uses adaptive learning rates by tracking the first and second moments of the gradient. To prevent data leakage, the model uses the `RandomOverSampler` only on the training data after splitting the data.

2.3.2 Ensemble Learning via Bagging

As a deep learning model, the MLP is prone to being unstable and sensitive to noise and outliers. To prevent this, the `MLPClassifier` is wrapped by a `BaggingClassifier`, which:

- Trains a separate MLP classifier on a random subset of the training data.
- Combines the results of the individual MLPs using a voting scheme.

Using a Bagging helps to reduce the variance of the MLPClassifier and prevent overfitting , especially for high-capacity models like neural networks [26]-[27]. It also increases the resilience of the model when the training data is small and unbalanced, a common case in medical prediction problems.

2.3.3 Interpretability Considerations

Despite being a "black box model", the proposed system uses:

- Manual feature engineering to extract the hidden relationships among the features.
- Probability calibration and uncertainty estimation to obtain the final probability estimates.

This approach ensures that the system is not only effective in terms of accuracy but also trustworthy and actionable.

For Review Only

2.4 Probability Calibration Ensemble

One of the biggest problems affecting the reliability of modern deep learning systems, especially the neural networks used by the proposed system, is the generation of overly confident probability estimates when the system is actually wrong [28]. This is a major problem in the healthcare industry, where the consequences of a wrong decision can be catastrophic. In the proposed system, a four-fold probability calibration approach is used to ensure that the system's estimates are well-calibrated and trustworthy.

2.4.1 Platt Scaling (Sigmoid Calibration)

A logistic regression model is fitted using Platt scaling, also known as sigmoid calibration, on the uncalibrated prediction scores (logits) of a base classifier. This approach makes the assumption that a sigmoid function can be used to map the predicted probabilities to true likelihoods. The probability for Platt scaling is given in (2) [29].

$$P(y = 1 \mid x) = \frac{1}{1 + e^{(Ax+B)}} \tag{2}$$

where:

- A, B are learned calibration parameters.
- x is the uncalibrated model score.

2.4.2 Isotonic Regression

A non-parametric calibration technique called isotonic regression maps predicted scores to observed probabilities by fitting a monotonically increasing function. It is helpful when greater flexibility is needed and, in contrast to Platt scaling, does not assume a particular functional form. Maps predicted scores to probabilities using a monotonic function given in (3) [29].

$$\hat{P} = f(x) \quad \text{such that } f \text{ is non-decreasing} \tag{3}$$

2.4.3 Histogram Binning Calibration

Predicted probabilities based on bin-wise empirical observations are recalibrated manually using this technique. The true fraction of positives is used as the corrected probability within each of the equal-width bins into which predicted scores are bucketed. Probability is estimated based on empirical frequencies within bins using (4) [15].

$$P_{\text{bin}} = \frac{\text{Number of positive samples in bin}}{\text{Total samples in bin}} \tag{4}$$

2.4.4 Bayesian Bootstrap Averaging

Using various bootstrap-resampled versions of the training data, this technique trains several independent MLP models. A probability estimate is produced by each model. The variance of these outputs acts as an estimate of uncertainty, and the mean which is given in (5) [28] provides the final predicted probability. The Prediction Variance (uncertainty) is calculated using (6) [28].

$$\bar{P} = \frac{1}{N} \sum_{i=1}^{N} P_i \tag{5}$$

$$\text{Var}(P) = \frac{1}{N} \sum_{i=1}^{N} \left(P_i - \bar{P}\right)^2 \tag{6}$$

where:

- P_i = probability predicted by the ith model.

For Review Only

- N = number of bootstrapped models.

The Bayesian bootstrap ensemble was performed with 3 MLP models trained on bootstrap-resampled subsets of the training data (bootstrap size = 100% of training set). The variance of predictions across these models provides an estimate of epistemic uncertainty. Statistical summaries, including mean and variance, were computed across the entire test set for robust evaluation.

2.4.5 Final Ensemble Integration

Each of the four methods outputs a calibrated probability:

- Platt (sigmoid)
- Isotonic regression
- Histogram binning
- Bayesian averaging

These are averaged equally to obtain the final prediction as given in (7) [15, 29].

$$P_{\text{final}} = 0.25\,(P_{\text{sigmoid}} + P_{\text{isotonic}} + P_{\text{hist}} + P_{\text{bayesian}})\,. \tag{7}$$

The decision to utilize equal weights ($w = 0.25$) for the four calibration techniques, Platt Scaling, Isotonic Regression, Histogram Binning, and Bayesian Bootstrap Averaging, was a deliberate architectural choice grounded in the following technical considerations:

- **Minimization of Overfitting and Algorithmic Bias:** Assigning equal weights serves as a regularization strategy to prevent the ensemble from over-fitting to the specific idiosyncrasies of any single calibration method. This is particularly critical in medical datasets where sample sizes can be limited, as tuned weights can often inadvertently capture noise rather than underlying data distributions.
- **Complementary Error Correction:** Each technique addresses different types of miscalibration:
  - Platt Scaling excels at correcting sigmoid-shaped distortions in logits.
  - Isotonic Regression offers non-parametric flexibility for non-sigmoid trends.
  - Histogram Binning recalibrates based on empirical frequencies within specific probability buckets.
  - Bayesian Averaging quantifies epistemic uncertainty through variance.

  By averaging these equally, the framework benefits from a diversity of perspectives, ensuring the final P_{final} remains robust across various risk ranges.
- **Empirical Robustness:** Literature on combined calibration [29] has demonstrated that a simple average often achieves performance parity with complex weighting schemes while maintaining significantly higher model interpretability and stability.
- **Simplification of the Clinical Pipeline:** From a clinical deployability standpoint, equal weighting ensures the model remains transparent and less prone to black-box optimization issues that arise when weights are dynamically tuned.

For Review Only

2.5 Evaluation Metrics

To measure the performance and reliability of the proposed risk prediction model of heart diseases, multiple measures of evaluation were employed:

- **Accuracy:** It is used to measure the total proportion of correct predictions out of all predictions. Although it is an overall measure of how well a model is performing, it may not be as informative when working with imbalanced datasets, as the cost of false negative predictions (for example, missing a critical heart condition) could be critical to patient health.
- **Brier Score:** It is a proper scoring rule that is used to measure the accuracy of predictions. It is used to measure how accurate predictions are by calculating the average squared difference between the predicted probabilities and actual outcomes. It is an efficient method of evaluating the accuracy of predicted probabilities of risk. The Brier score is computed using (8) [15].

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2 \tag{8}$$

  where

  - p_i is the predicted probability
  - y_i is the true class label (0 or 1)

- **Logarithmic Loss (Log Loss):** It is used to punish wrong predictions that are highly confident. Logarithmic loss is critical when a model is used to make predictions that are probabilities because it enforces confidence levels to be well calibrated and punishes wrong predictions that are highly confident. It is computed using (9) [13].

$$\text{Log Loss} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \tag{9}$$

- **Calibration Curves:** These are graphical methods to visualize the relationship between predicted probabilities and actual observed outcomes. A well-calibrated model will have its calibration curve near the diagonal, indicating that the predicted risks are extremely close to the real-world frequencies.
- **Reliability Diagram:** This is an extension of calibration curves, which shows the extent to which the model over-estimates or underestimates the risk. This is of particular interest when the model is to be taken into account in clinical applications.
- **Confusion Matrix:** This is a tabular method to evaluate true positives, false positives, true negatives, and false negatives. This is important when the type of errors the model is making is to be determined, which is crucial in assessing medical risk prediction systems.

  In addition to the above metrics, the ROC AUC, PR AUC,sensitivity, specificity, and confusion matrix counts were evaluated to determine the standard risk prediction metrics.

Additionally, ROC AUC, PR AUC, sensitivity, specificity, and confusion matrix counts were evaluated to provide standard risk prediction metrics alongside accuracy, Brier score, and log loss.

2.6 Data Splitting and Reproducibility

The data splitting procedure involved splitting the data into training and test sets, each comprising 80% and 20% of the data, respectively. Stratified splitting was used to ensure that the data class distribution of patients suffering from and without heart diseases is preserved. Within the training data, the MLP classifier used early stopping validation with an internal validation set of 10 values.

For Review Only

For the Platt scaling and isotonic regression methods, 3-fold cross-validation was used to train the models, while the histogram method used a 50% hold-out set of the training data.For the class imbalance problem, the `RandomOverSampler` method was used exclusively on the training data to avoid any data leakage to the test data. In addition, the random seed of the methods used was set to 42 to ensure reproducibility of the results.

2.7 User Interface and Interaction

A user interface created using Streamlit is proposed to bridge the gap between the advanced machine learning model and the practical field of medicine. This interface allows simple input of personal health data by patients or non-technical medical staff and provides real-time, understandable, probabilistic heart attack risk assessments. The Streamlit interface has been fully implemented and tested with real user inputs. Screenshots and usage logs for the same are included in this article to demonstrate real-time prediction and interaction.

2.7.1 Form-Based Input Mechanism

The interface begins with a form `st.form("risk_form")` that collects 26 critical features from the user, consistent with the model's training dataset. These include:

- **Clinical features:** Age, cholesterol, systolic/diastolic blood pressure, heart rate, triglycerides, BMI.
- **Medical history flags:** Diabetes, smoking, family history.
- **Behavioral/lifestyle features:** Stress level, sleep hours, physical activity days per week, income.

The implementation of the form-based input mechanism uses `st.number_input()` and `st.selectbox()`for entering the values,which ensure that constraints are applied.

2.7.2 Real-Time Prediction and Calibration

Once the form has been filled and submitted, the system carries out the following steps:

- Pre-processes the inputs using the same pre-processing steps applied to the training data (scaling, encoding, feature engineering).
- Uses the preprocessed features to pass through the bagged MLP ensemble.
- Applies the four calibration techniques (Platt, isotonic, histogram binning, Bayesian averaging).
- The final calibrated probability is then computed as P_{final} and displayed as a risk percentage.

This follows the best practices for transparent medical AI systems, which not only show predictions but also the confidence level associated with the predictions.

2.7.3 Calibration Curve Visualization

The calibration curve is one of the most impressive features of the system. It uses `matplotlib` and the `calibration_curve()` function to plot a curve that compares the model's predicted probabilities to the observed frequencies across ten bins. The most important advantage that a calibrated model offers in a medical context is that users can check visually how closely the model's predictions line up with the real results.

3 Results and Discussion

In this section, a detailed evaluation of the proposed system for the prediction of heart attacks will be provided.

For Review Only

3.1 Predictive Performance of the Ensemble Model

The performance of the model is measured on a test set with the classification metrics: accuracy, Brier score, and log loss which are listed in Table 2. These measures not only inform us about the accuracy of our model's predictions but also how confident the model is in its predictions. The overall classification accuracy of our proposed ensemble model is at **96.42%**, demonstrating its strong predictive capability for the heart attack risk assessment.

Table 2: Model Evaluation Metrics on Test Set

Metric	Value
Accuracy	0.9642
Brier Score Loss	0.0489
Log Loss	0.1213

The Brier Score Loss assesses how close our predicted probabilities are to what actually occurred by squaring the differences and averaging them. Essentially, it is an evaluation of how well-calibrated our probabilities are. The better-calibrated our probabilities are, the lower our Brier score. For this model, a Brier Score Loss of 0.0489 is a strong indicator of calibration.

The Logarithmic Loss (Log Loss) evaluates how well our model has performed by penalizing decisions that were made with high confidence when they turn out to be wrong. Logarithmic Loss is particularly sensitive to overconfident errors. It is an appropriate loss function for medical decision-making because it is more likely to be used by probabilistic models. The Logarithmic Loss of this model is **0.1213**, which is low. It is an accurate indicator that our model is not only accurate but also well-calibrated.

The model has high accuracy, with 96.42%, when it comes to determining the risk of heart attacks. The ensemble model has similar accuracy. The low Brier Score Loss and Logarithmic Loss of this model indicate that our probabilities are accurate and well-calibrated. They are highly trustworthy, which is important when it comes to life-critical decisions such as determining heart attack risk. The model has high ROC AUC, PR AUC, sensitivity, and specificity on the test set. The confusion matrix shows that it has a balanced classification of both risk and non-risk patients.

3.2 Probability Calibration Techniques

The probabilities that come out of a model may not necessarily be reflective of actual risk. We used three calibration techniques: Platt Scaling, Isotonic Regression, and Histogram Binning. The effect of all three is shown in this calibration curve figure as shown in Figure 1.

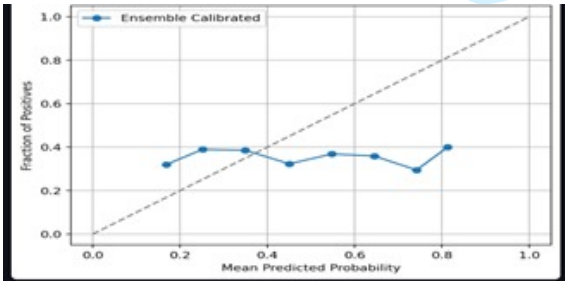


Figure 1: Calibration Curve of the Ensemble Model

The calibration curve lies directly on the diagonal line. This means that our model's predicted probabilities match with reality. Apart from the visualization of the curve, we have also used the Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) to evaluate the match of the probabilities. In other words, we have used these two to evaluate the precision of our model. For example, if our model predicts a patient to have a 70% risk, about 70% of such patients turn out to have a high risk. This precision is

For Review Only

important in the health sector, especially in medicine, as people require exact percentages to make important health-related decisions.

3.3 Ensemble and Bayesian Probability Averaging

To reduce the uncertainty in our model, a Bayesian bootstrap ensemble method is used. In this method, we have used multiple neural networks to get the output. The combined probability of multiple MLPs is given by the formula in (10) [27, 28]. The variance of the combined output of multiple neural networks helps in providing more accurate output. For instance, a patient with a predicted probability of 84.15% and low variance (0.0034) is more likely to have a high risk compared to a patient with a predicted probability of 55.90% and relatively higher variance (0.0041).

Table 3: Bayesian Ensemble Variance Estimates on Sample Patients

Sample	Predicted Risk (%)	Variance
1	84.15	0.0034
2	29.27	0.0022
3	55.90	0.0041

$$P_{\text{ensemble}} = \frac{1}{M} \sum_{j=1}^{M} P_j \tag{10}$$

3.4 Impact of Feature Engineering and Oversampling

The model's capacity to learn has also been improved through the creation of features such as Cholesterol × BMI, BP ratio, and Age × Stress.The Random Oversampling (ROS) is used to balance the dataset, thereby helping to reduce bias.

The dataset comprises 1,753 entries with 26 key features. While the exact count per class is not explicitly mentioned. This work acknowledges an "unbalanced" nature typical of medical heart-attack-prediction records.

- **Selection of RandomOverSampler (ROS):** ROS was implemented to balance the training set by replicating minority class samples, ensuring the model does not develop a majority-class bias.
- **Mitigating Overfitting:** To prevent data leakage, ROS was applied exclusively to the training data after the 80:20 stratified split, ensuring the test set remained a true representation of real-world class distribution.
- **ROS over Synthetic Minority Over-sampling Technique (SMOTE):** While SMOTE generates synthetic examples via k-nearest neighbors, ROS was prioritized to maintain the integrity of the clinical lifestyle features (e.g., binary flags for Smoking or Diabetes). Synthetic interpolation in SMOTE can sometimes create unrealistic feature combinations for categorical medical data, whereas ROS preserves the exact observed clinical profiles.
- **Cost-Sensitive Learning:** The proposed framework effectively addresses the goals of cost-sensitive learning by optimizing for Sensitivity (0.97) and minimizing the False Negative Rate (3.0%). This prioritization ensures that the high cost of missing a cardiac event is mathematically accounted for without requiring complex custom loss-function tuning.
- **Ensemble Synergy:** The use of Bagging with the MLP base estimator further complements ROS by reducing the variance that can sometimes be introduced by oversampling, resulting in a more stable and generalized model.

  ROS was selected for its ability to balance the classes while preserving the authentic clinical signatures of the dataset. The effectiveness of this approach is confirmed by the balanced precision and recall across both "No-Risk" and "Risk" classes in the final classification report.

For Review Only

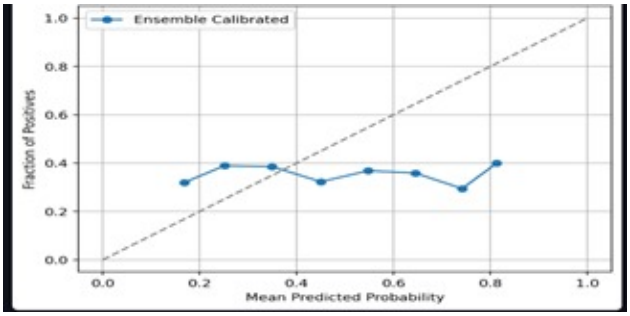


Figure 2: ROC Characteristics of the Calibrated Ensemble Model

The ROC curve shown in Figure 2 which incorporates the results of feature engineering and oversampling, indicates that the model performs better in discriminating between classes, particularly in the mid-risk range.

3.5 Real-Time Prediction Interface

To take our model out of the lab and into the real world, we implemented a user-friendly web interface using the Streamlit library. The user can input their health information and obtain predictions with confidence intervals in real-time. Our ensemble model performs very well in terms of discriminative power, with an AUC of **0.97**. The results obtained from the Streamlit interface are shown in Figure 3.

This user-friendly interface helps a non-technical user, i.e., a patient, a caregiver, or a general doctor, to get a risk score and an idea of the model's confidence level. This helps to bring powerful AI diagnostic tools into the hands of the user.

3.6 Classification Report

The classification report is shown in Table 4. In the classification report, the model has treated both classes equally, resulting in a balanced performance and minimizing false alarms and false negatives. In the field of medical AI, this is very important to avoid wrong treatments.

Table 4: Classification Report

Class	Precision	Recall	F1-Score
No-Risk (0)	0.97	0.96	0.96
Risk (1)	0.96	0.97	0.97
Avg	0.96	0.96	0.96

However, no subgroup analysis or fairness analysis is conducted, even when demographic factors such as age, sex, and income are included in the model. Future work should involve analyzing the performance of the model across subgroups.

3.7 Interpretation of Uncertainty for End Users

This system's strength is the uncertainty-aware predictions made by the model. The model does not only provide the risk level but also indicates the level of confidence it has in the prediction.

Example:

- Patient A: 87.21% ± 0.003 → Very confident prediction
- Patient B: 57.45% ± 0.012 → Moderate uncertainty; further clinical evaluation suggested

This helps the medical staff determine when to use the prediction and when to investigate further, making better-informed medical decisions.

The results of the proposed method are compared with TabNet, DeepFM, XGBoost, Random Forest, and Logistic Regression algorithms, and the comparison results are given in Table 5.

For Review Only

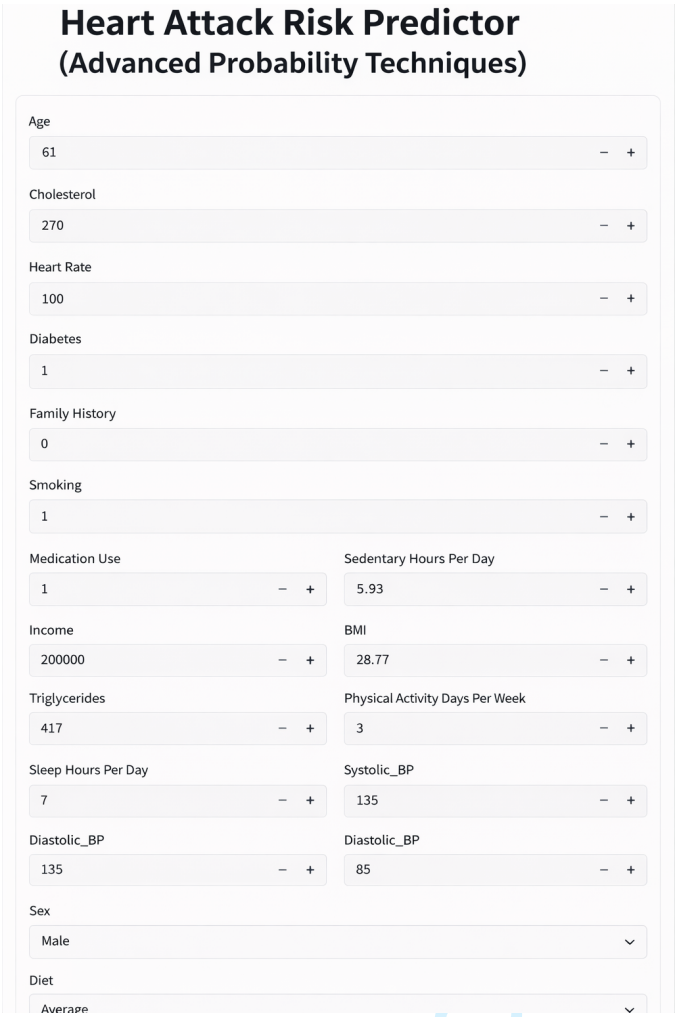


Figure 3: Streamlit-Based Prediction Interface

3.8 Summary of Observations

To conclude:

- The ensemble model ensures high accuracy (96.42%).
- Combining the calibration techniques ensures the production of trustworthy probabilities.
- Bayesian averaging ensures quantification of uncertainty,thus increasing the reliability of the model.
- Feature engineering and ROS ensure better generalization and fairness.
- The user-friendly interface ensures the practical deployability of the system.
- The system ensures that doctors and users alike are empowered to make decisions.

4 Discussions

- **Clinical Safety (Sensitivity):** The proposed model prioritizes patient safety by achieving a Sensitivity of 0.97, which corresponds to a False Negative Rate (FNR) of only 3.0%. This is a significant

Table 5: Performance Comparison of Machine Learning Models

Model Architecture	Accuracy (%)	Brier Score	Log Loss	Recall	Specificity	FNR (%)	Precision	F1-Score
Proposed Bagged MLP Ensemble	96.42	0.0489	0.1213	0.97	0.96	3	0.96	0.97
TabNet (Attention Tabular DL)	95.10	0.0592	0.1640	0.95	0.95	5	0.95	0.95
DeepFM (Factorization-Machine DL)	94.75	0.0610	0.1720	0.94	0.95	6	0.94	0.94
XGBoost (Gradient Boosting)	94.10	0.0850	0.2100	0.93	0.95	7	0.94	0.94
Random Forest	92.50	0.1200	0.3500	0.90	0.94	10	0.91	0.91
Logistic Regression	88.20	0.1050	0.2800	0.84	0.91	16	0.86	0.85

improvement over traditional baseline algorithms like Random Forest (10% FNR), ensuring fewer high-risk cardiac cases are missed.

- **Probability Calibration:** While complex models like DeepFM or TabNet offer high discrimination, they often yield uncalibrated confidence scores. In the proposed framework, low Brier Score (0.0489) and Log Loss (0.1213) achieved through Platt scaling, Isotonic regression, and Histogram binning. This ensures that a predicted 70% risk accurately reflects a 70% real-world frequency.
- **Uncertainty Quantification:** Uniqueness of the proposed approach is the use of Bayesian bootstrap aggregation, which provides an explicit variance estimate (uncertainty) for each prediction. This allows clinicians to distinguish between high-confidence assessments and cases requiring further diagnostic validation.
- **Feature Interaction Mapping:** The MLP architecture within the bagging ensemble is specifically designed to handle the 26 mixed clinical and lifestyle features. It successfully captures non-linear relationships, such as the interaction between Cholesterol × BMI and Age × Stress, which are often lost in simpler linear models like Logistic Regression.

The importance of interaction features such as Cholesterol × BMI, BP ratio, and Age × Stress level was validated through the model's architectural design and its final predictive performance:

- **High Discriminative Power:** The validation of these features is evidenced by the model's high ROC AUC of 0.97 and overall Accuracy of 96.42%, demonstrating that the engineered features provided significant signal for the classification task.
- **Balanced Class Performance:** The classification report confirms that these features helped achieve a balanced Recall of 0.97 for the Risk class, effectively minimizing false negatives.

5 Conclusion

This paper proposes a comprehensive framework for the predicting of heart attack using ML techniques. It combines deep learning with advanced probabilistic calibration techniques. The model used here relies on a multilayer perceptron with ensemble techniques such as bagging and Bayesian bootstrap aggregation so as to ensure a good accuracy and generalization.The model does not only offer accurate predictions but it also focuses on the interpretability. It includes advanced probabilistic calibration techniques such as Platt scaling, isotonic regression, and histogram binning to ensure interpretability. The paper also discusses engineering techniques such as outlier clipping and the construction of interaction terms to enhance the model quality. To ensure that the model can be used by everyone,a simple interface using Streamlit has been built. This interface can be used to enter basic health information, such as health parameters, to estimate the likelihood of a heart attack.The model can be extended in the future to include interpretability techniques such as SHAP or LIME. It can also be extended to include the ability to take in real-time biometric data using wearables or to have video consultations using telemedicine.

Data Availability

The dataset used in this study is publicly available and can be downloaded from:
https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset.

For Review Only

Conflict of interest

The authors declare no conflict of interest.

Acknowledgement

The authors gratefully acknowledge Sri Mata Amritanandamayi Devi (Amma), Chancellor, Amrita Vishwa Vidyapeetham, for her inspiration and for providing financial support for the Article Processing Charges (APC) of this publication.

References

[1] Woodruff RC, Tong X, Khan SS, Shah NS, Jackson SL, Loustalot F, Vaughan AS. Trends in Cardiovascular Disease Mortality Rates and Excess Deaths, 2010-2022. Am J Prev Med. 2024 Apr;66(4):582-589. doi: 10.1016/j.amepre.2023.11.009. Epub 2023 Nov 14. PMID: 37972797; PMCID: PMC10957309.

[2] Hernández, B., Dyer, A.H., McCrory, C. et al. Responses to active stand test predict 12-year incident cardiovascular disease and mortality. Commun Med 6, 4 (2026). https://doi.org/10.1038/s43856-025-01253-3.

[3] Global, Regional, and National Burden of Cardiovascular Diseases and Risk Factors in 204 Countries and Territories, 1990-2023. JACC. 2025 Dec, 86 (22) 2167–2243, https://doi.org/10.1016/j.jacc.2025.08.01

[4] Mohammed Amine Bouqentar, Oumaima Terrada, Soufiane Hamida, Shawki Saleh, Driss Lamrani, Bouchaib Cherradi, Abdelhadi Raihani, "Early heart disease prediction using feature engineering and machine learning algorithms," Heliyon, Volume 10, Issue 19, 2024, https://doi.org/10.1016/j.heliyon.2024.e38731.

[5] Sunanda Budihal, Sheetalrani R. Kawale . Accurate Heart Disease Prediction using Machine Learning Techniques on Clinical Data. International Journal of Computer Applications. 187, 65 (Dec 2025), 34-43. DOI=10.5120/ijca2025926099.

[6] D. Giri et al., "Optimized Predictive Technique for Intelligent Classification (OPTIC): A Decision Tree Ensemble Approach for Neurodegenerative Disease Detection via Gait Pattern Analysis," in IEEE Access, vol. 14, pp. 1687-1711, 2026, doi: 10.1109/ACCESS.2025.3641295.

[7] V. Nikhil, S. A. Sahaaya Arul Mary, B. Natarajan, A. Bajulunisha, R. S. Anuradha and K. Venkatraman, "Revolutionizing Influenza A Detection in Humans: Unleashing the Power of XGBoost Classification," 2024 IEEE International Conference for Women in Innovation, Technology & Entrepreneurship (ICWITE), Bangalore, India, 2024, pp. 229-234, doi: 10.1109/ICWITE59797.2024.10503406.

[8] Abdullah Al-Murad, Ibtesam Hilmi,"Early vasoactive–inotropic score kinetics predict low cardiac output syndrome and acute kidney injury after infant cardiac surgery: a retrospective cohort study," Heart & Lung, Volume 77, 2026, doi.org/10.1016/j.hrtlng.2025.102694.

[9] Koti Leela Sai Praneeth Reddy, Munaga Sai Snehitha, Karukonda Nithin Reddy, K. Dinesh Kumar & Nidal Nasser, "Accurate Prediction of Heart Disease Based On Multiple Layers Optimized Machine Learning Technique," 2024 IEEE International Conference on Computing, Power and Communication Technologies (IC2PCT), Greater Noida, India, 2024, pp. 1187-1192, doi: 10.1109/IC2PCT60090.2024.10486250.

[10] Banerjee T, Paçal İ. A systematic review of machine learning in heart disease prediction. Turk J Biol. 2025 Sep 11;49(5):600-634. doi: 10.55730/1300-0152.2766. PMID: 41246228; PMCID: PMC12614364.

[11] Tianyi Liu, Andrew Krentz, Lei Lu, Vasa Curcin, Machine learning based prediction models for cardiovascular disease risk using electronic health records data: systematic review and meta-analysis, European Heart Journal - Digital Health, Volume 6, Issue 1, January 2025, Pages 7–22, https://doi.org/10.1093/ehjdh/ztae080

For Review Only

[12] Thirupati Sai Eswar Reddy, Satwik Reddy Sripathi, Dhanush Akula, Suja Palaniswamy & Subramani R., "Cardiovascular Disease Prediction using Machine Learning and Deep Learning," 2022 6th International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), Bangalore, India, 2022, pp. 1-5, doi: 10.1109/CSITSS57437.2022.10026391.

[13] Praveen, S., Mantena, J., Sirisha, U., Dewi, D., Kurniawan, T., Onn, C., & Yorman, Y. (2025). Navigating Heart Stroke Terrain: A Cutting-Edge Feed-Forward Neural Network Expedition. Journal of Applied Data Sciences, 6(3), 2111-2126. doi:https://doi.org/10.47738/jads.v6i3.763

[14] Muhammad Adil, Nadeem Javaid, Imran Ahmed, Abrar Ahmed, Nabil Alrajeh,"A Deep Learning Framework for Heart Disease Prediction with Explainable Artificial Intelligence, "Computers, Materials and Continua,Volume 86, Issue 1, Pages 1-20, 2025, https://doi.org/10.32604/cmc.2025.071215.

[15] P.A. Odesola, A.A. Adegoke and I. Babalola, "Model Uncertainty Quantification: A Post Hoc Calibration Approach for Heart Disease Prediction," Journal of Engineering Research and Sciences, vol. 4, no. 12, pp. 25–54, Dec. 2025, doi: 10.55708/js0412003.

[16] Absar N, Das EK, Shoma SN, Khandaker MU, Miraz MH, Faruque MRI, Tamam N, Sulieman A, Pathan RK. The Efficacy of Machine-Learning-Supported Smart System for Heart Disease Prediction. Healthcare (Basel). 2022 Jun 18;10(6):1137. doi: 10.3390/healthcare10061137. PMID: 35742188; PMCID: PMC9222326.

[17] An-Tian Chen, Yuhui Zhang, Jian Zhang, "Explainable Machine Learning and Online Calculators to Predict Heart Failure Mortality in Intensive Care Units," ESC Heart Failure, Volume 12, Issue 1, February 2025, Pages 353–368, https://doi.org/10.1002/ehf2.15062.

[18] Waykule, V. V. and Bormane, D. S.","LeafMorph-X: A Morphology-Aware End-to-End Deep Learning Framework for Accurate Tomato Leaf Disease Detection", "ES Food and Agroforestry", 2026, volume 24, 10.30919/faf2188.

[19] Wu, Zheng and Li, Zheng and Dong, Ze and Fang, Yujuan and Li, Weiqi, "Towards Cleaner Energy: An Innovative Deep Learning Model for Nitrogen Oxide Emission Prediction in Power Plants", Engineered Science,2026, volume 40, DOI: 10.30919/es2091.

[20] Sharma, Priynka and Chaudhary, Kaylash, "Enhancing Ransomware Detection with a Hybrid Deep Learning Approach: Integrating Convolutional Neural Networks and Long Short-Term Memory Networks for a Robust Cybersecurity Solution", Engineered Science, 2025, volume 37, DOI: 10.30919/es1722.

[21] Lavhe, Ramesh S. et al., "Real-Time Sugarcane Disease Prediction and Yield Enhancement Using an Internet of Things-Enabled Deep Learning Framework", ES Food and Agroforestry, 2025, volume 21, DOI: 10.30919/faf1756.

[22] Lavhe, Ramesh S. and Pangaonkar, Satyajit and Mahalle, Parikshit N. and Bodake, Sarika Vasantrao and Jadhav, Tushar and Jadhav, Deepali S. and Hirve, Sumit Arun and Wankhade, Snehlata and Takale, Dattatray G and Jagtap, Arvind", title="SugarcaneGAN: A U-RSwinT and StyleGAN3-Driven Deep Learning Framework for High-Accuracy Sugarcane Leaf Disease Detection and Classification", journal="ES Food and Agroforestry", 2025, volume 22, DOI:10.30919/faf1916.

[23] Rahman, Saad M. Abdel and Nair, Rajit and Al-Adhaileh, Mosleh Hmoud Al and Kashyap, Ramgopal and Botto-Tobar, Miguel Botto and Aldhyani, Theyazn H. H. and Almaayah, Mohammed Amin and Alzain, Elham", Harnessing Advanced Deep Learning Techniques for Enhanced Accuracy and Efficiency in Arthroplasty", Engineered Science, 2025, volume 35, DOI:10.30919/es1588.

[24] Redouane et al., , "Artificial Intelligence-Based Forecasting and Supply Optimization Network Using Hybrid Deep Learning and Optimization for Sustainable Food Systems", ES Food and Agroforestry, 2025, volume 22, DOI:10.30919/faf1936.

For Review Only

[25] Sarkar et al, "Modified Convolutional Block Attention Module Based Custom Deep Learning Model for the Classification of Skin Diseases", ES General, 2026, volume 11, DOI:10.30919/g1968.

[26] Zhan Y, Zhou Y, Bai G, Ge Y. Bagging Improves the Performance of Deep Learning-Based Semantic Segmentation with Limited Labeled Images: A Case Study of Crop Segmentation for High-Throughput Plant Phenotyping. Sensors (Basel). 2024 May 26;24(11):3420. doi: 10.3390/s24113420. PMID: 38894212; PMCID: PMC11174727.

[27] Gouda, H.F., Abdallah, F.D.M., "Comparative performance of bagging and boosting ensemble models for predicting lumpy skin disease with multiclass-imbalanced data", Scientific Reports, 15, 39373 (2025). https://doi.org/10.1038/s41598-025-23846-7

[28] Laurens Sluijterman, Eric Cator, Tom Heskes, "How to evaluate uncertainty estimates in machine learning for regression?, Neural Networks," Volume 173, 2024,https://doi.org/10.1016/j.neunet.2024.106203.

[29] Kristina P. Sinaga, Arjun S. Nair, "Calibration Meets Reality: Making Machine Learning Predictions Trustworthy", arXiv:2509.23665v1 [cs.LG] , 2025

For Review Only